Neural Networks

Multi-Layer Perceptron (MLP) Neural Networks

Slides are partially adapted from notes by Vincent Cheung, Kevin Cannons and William Kaiser

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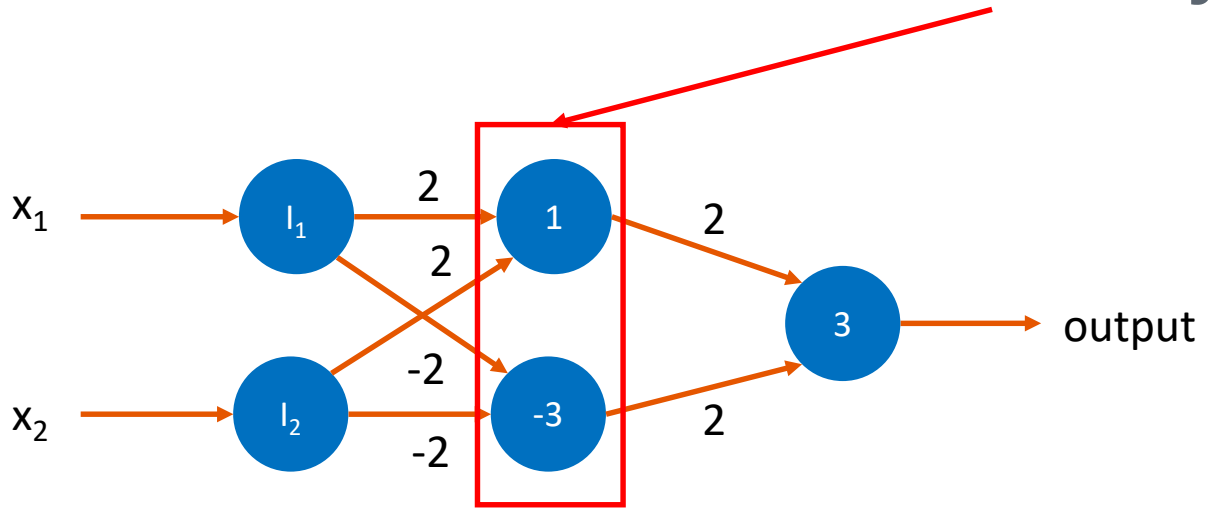
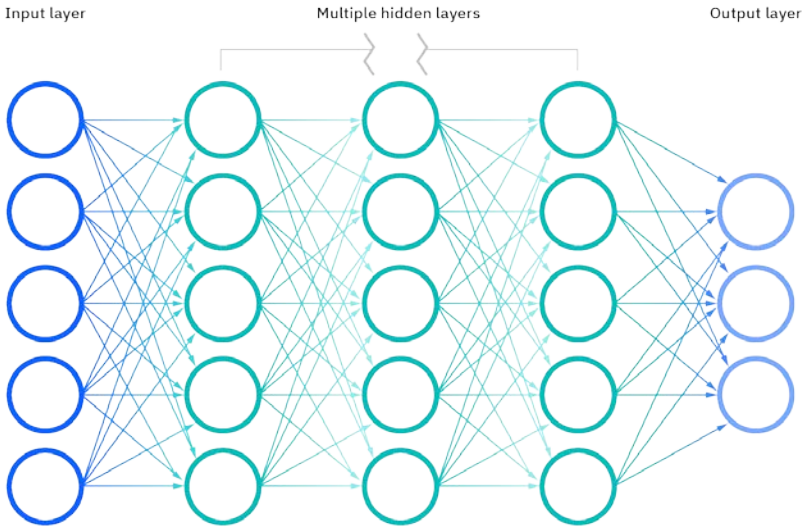
Recall: XOR Gate

- The XOR Gate can be implemented with a new neural network structure that includes additional neurons that are between the network inputs and network output.
- The new structure includes a “Hidden Layer”.



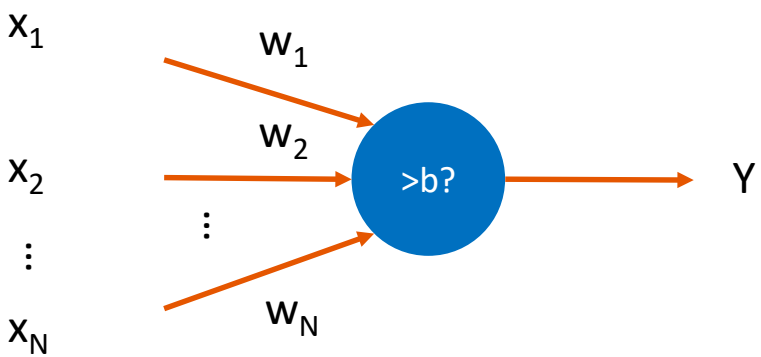
Multi-Layer Perceptron Neural Networks

- Multi-Layer Perceptron (MLP) are artificial neural networks that have one or more hidden layers.
- Can be used to approximate more complex functions.

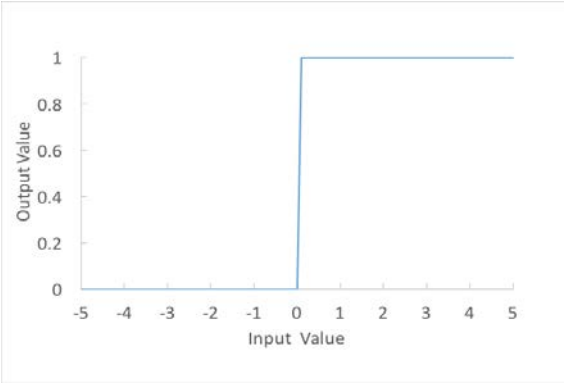


Activation Functions

Rewrite the condition to determine if a perceptron's output is activated:



$y = 1$ if $x_1w_1 + x_2w_2 + \ldots + x_Nw_N - b > 0$
$y = 0$ if $x_1w_1 + x_2w_2 + \ldots + x_Nw_N - b \leq 0$



The perception uses a simple step function as the **activation function**.

Activation Functions (cont'd)

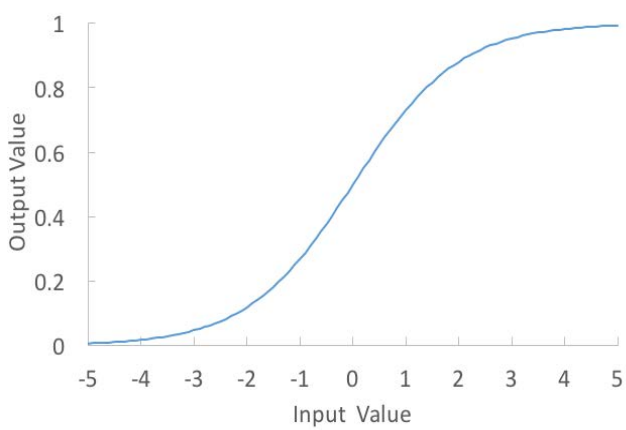
Drawback of a step activation function:

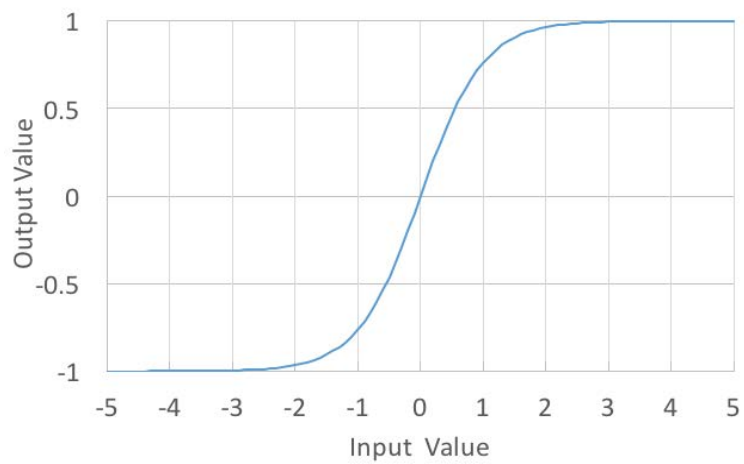
- A small change in an input value may cause the perceptron to change state from 0 to 1.

It is more desirable a perceptron responds in a continuous manner where a small change in input may lead to a small change in output.

Commonly Used Activation Functions

Commonly used activation functions for $y = \left(\sum_{j=1}^{N} w_j x_j\right) - b$



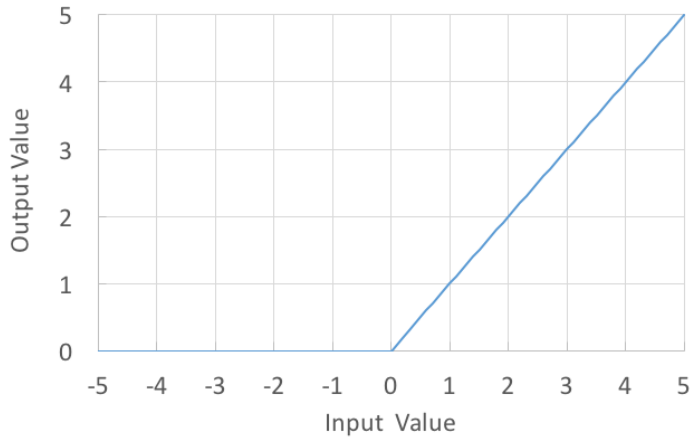


Sigmoid (logistic): $f(y) = \frac{1}{1+e^{-y}}$

Tanh: $f(y) = \frac{e^{y}-e^{-y}}{e^{y}+e^{-y}}$

Commonly Used Activation Functions (cont'd)

- **Commonly used activation functions for** $y = \left(\sum_{j=1}^{N} w_j x_j\right) - b$



Rectified Linear Unit (ReLU):

$$f(y) = \max(0, y)$$

Question

If a Rectified Linear Unit activation function $f(y) = \max(0, y)$ is used, what is the output given the input is -2?

a) -2

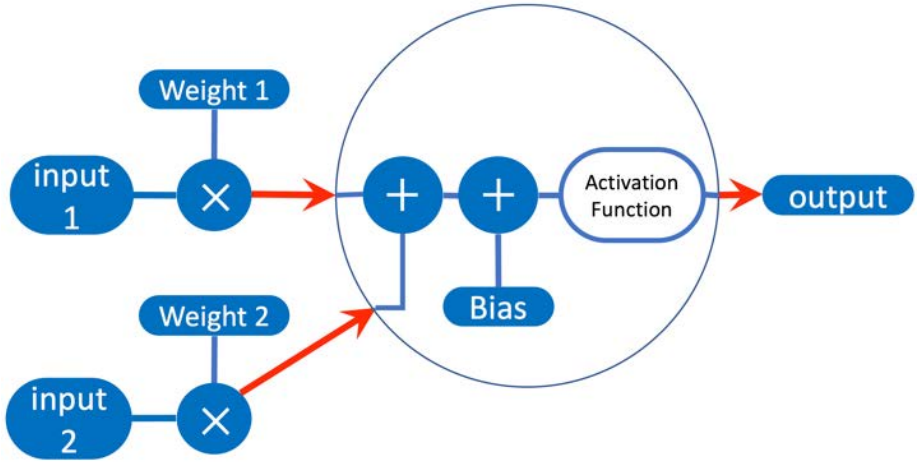
b) 0

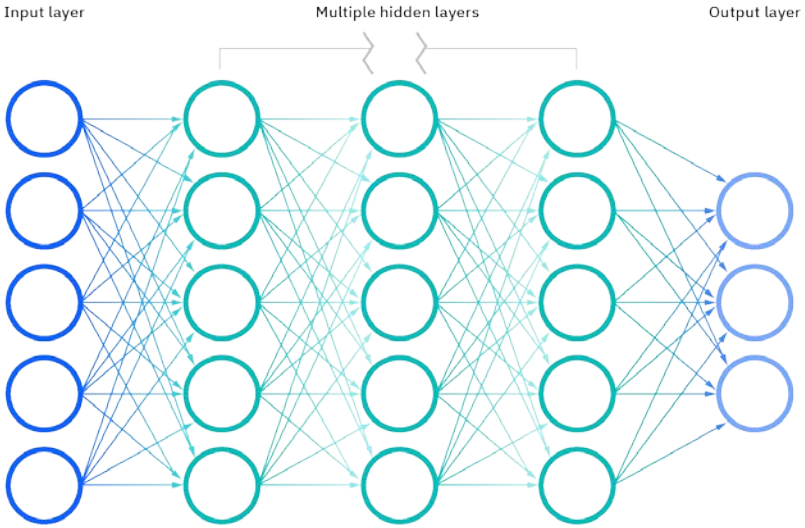
c) 2

How do MLP Neural Networks work?

The function of the entire neural network is simply the computation of the outputs of all the neurons.

- An entirely deterministic feedforward calculation





$$\text{Neuron Output} = f(y) = f\left(\left(\sum_{j=1}^{N} w_j x_j\right) + b\right)$$

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